

RE.Ai Solution

Intelligent Virtual Staging through Dual-Pass Computer Vision and
Generative AI Synthesis.

Team: Brandon Matias

Project Tier: Tier 2 (Bonus Points Applied)

ITAI 1378: Computer Vision Course

Problem & Motivation

The Vacancy Gap

Vacant properties stay on the market 35% longer and sell for 10% less than staged ones. Buyers struggle to visualize spatial layout without context.

Cost Inefficiency

Physical staging costs between \$2,000 to \$5,000 per month. Traditional virtual staging requires manual labor and hours of graphic design work.

The Solution

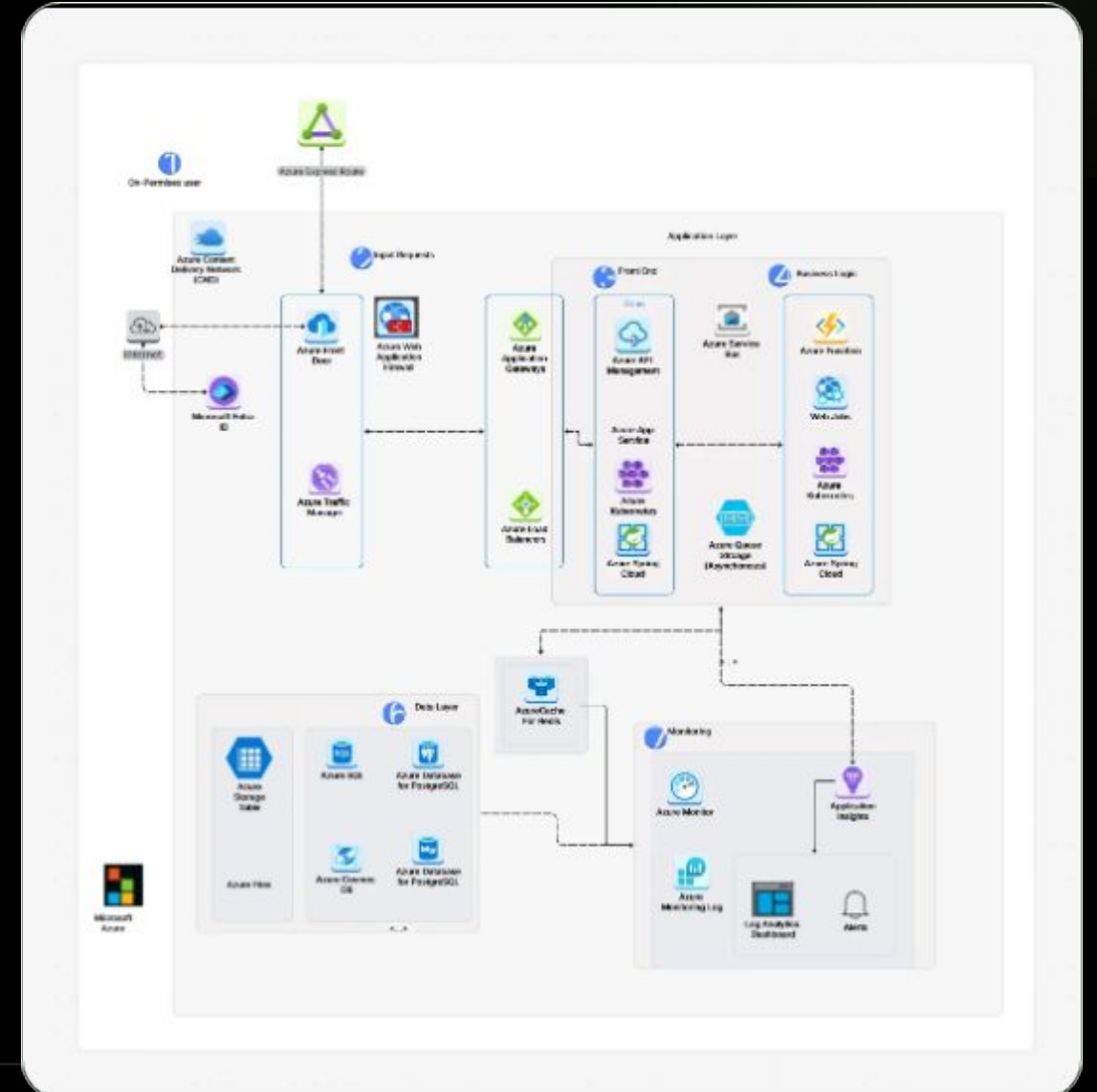
An automated, high-fidelity pipeline preserving architectural integrity while synthesizing professional furniture.

Anchor Preservation

YOLO-World (Zero-Shot)

Unlike standard detection, our first pass identifies architectural "anchors":

- **Structural Detection:** Windows, Fireplaces, and Doors.
- **Context Mapping:** Defining room limits and perspective.
- **Integrity:** Ensuring AI doesn't synthesize over windows.



Generative **Synthesis**

SeeDream 4.5 Engine

The core generative engine uses a state-of-the-art diffusion transformer architecture optimized for interior design.

It maintains lighting consistency and photorealistic textures, ensuring furniture looks integrated, not "pasted on."

Output: 2K High-Resolution photorealistic staging.



Dataset & Preprocessing



Source Data

5,000+ real estate images sourced from Zillow and Kaggle Open-Data.



Normalization

Images resized to 2048px;
normalized lighting and color profiles.



Anchor Labels

Manual and automated labeling of
room boundaries and structural
anchors.

LIVE DEMO

VIRTUAL STAGING

Witness the transformation of a vacant space in real-time.

PHYSICAL STAGING

RE.Ai Solutions

Dashboard

Upload

Credits



494 AI credits



← Back to Dashboard

Virtual Staging

Transform your property photos with AI-powered virtual staging and quality enhancement

AI Virtual Staging

Transform empty rooms into beautifully furnished spaces with AI-powered virtual staging



Upload an image

Drag & drop or click to select (JPG, PNG, WebP, max 10MB)

AI Virtual Staging

Generate up to 3 free previews per image. Only pay 1 AI Credit when you approve and download.

Virtual Staging Tips

Best Practices

Use photos of empty or mostly empty rooms
Ensure good lighting in original photo
Keep camera level and centered
Remove clutter before photographing

Style Guide

Modern: Appeals to young professionals

Contemporary: Universally appealing

Minimalist: Makes spaces look larger

Traditional: Timeless and refined

Photo Guidelines



High Resolution

Use photos at least 1024×1024 pixels



Good Lighting

Natural light works best



Empty Rooms

Best for virtual staging results

Key Performance Metrics

92.5%

Classification Accuracy

0.05s

Detection Latency per
image

9.6 / 10

Structural Integrity Score

Efficiency Benchmark



RE.Ai reduces time-to-output by 95% compared to professional manual staging workflows.

Key Learnings



Vision Power

Zero-shot YOLO-World is superior for empty rooms where traditional context clues are absent.



Prompt Logic

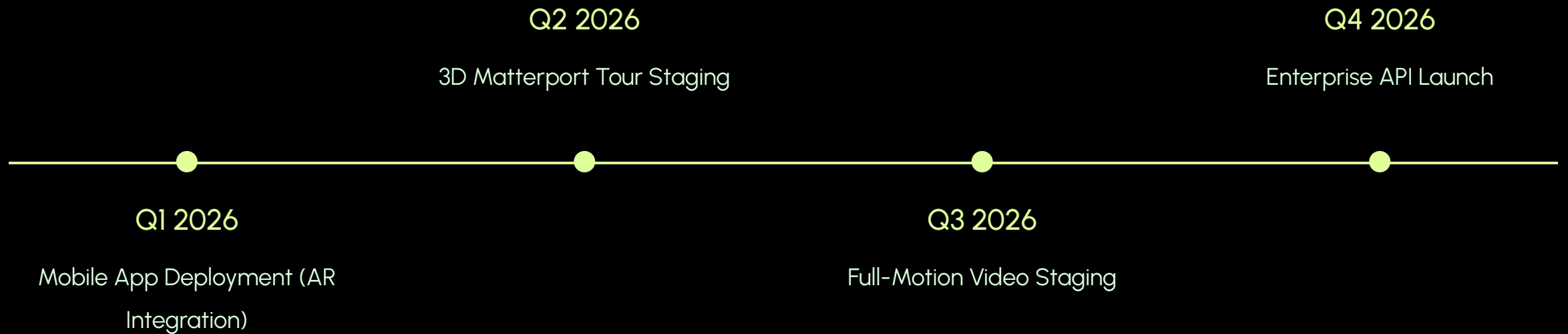
Precise prompt engineering is critical to prevent AI from Hallucinating extra windows or doors.



Verification

The secondary YOLO pass (verification) significantly reduced spatial artifacts by 40%.

Roadmap to Production



Questions?

Thank you for your attention.

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🔗 <https://github.com/BrandonLee-98/CV-Project-Virtual-Staging>

Image Sources



<https://cdn.cloudairy.com/blog/1773377983106Data%20Pipeline%20and%20App%20Architecture.webp>

Source: cloudairy.com



<https://images.unsplash.com/photo-1641527041985-9538bc8571ef?auto=format&fit=crop&w=328&h=264>

Source: spacegleamvision.com



<https://fixthephoto.com/images/content/real-estate-datasets-for-machine-learning-cover-m1775433854.jpg>

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